

2079

Bachelor of Science in Computer Science and Information Technology

Course Title: **Database Management System**

Code No: **CSC265** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What is Database Management System? Explain the DBMS architecture with diagram. What is data independence?
2. What is entities and its types? How do we reduce different types of entities and relationships of ER diagram into Tables? Explain with assuming different types of entities and relations.
3. What is normalization? Why normalization is required? Explain 1NF, 2NF and 3NF with example.

Section B

Attempt any Eight questions[8×5=40]

4. What do you mean by Schema and instance in DBMS? Explain both with example.
5. What is conflict schedule? Explain with example.
6. Retrieve the TName, and No_of_priod of teachers who teach in "ABC" school using Relational Algebra. TEACHER(TID, TName, TAddress, TQualification) SCHOOL (SID, SName, SAddress, SPhone) SCHOOL_TEACHER (SID, TID, No_of_period)
7. Explain aggregation with example.
8. Define functional dependencies. Explain 2nd normal form with example?
9. Explain Assertion and Triggers with example.
10. What is concurrency control? What are its advantages in DBMS?
11. What is Buffer Management in DBMS? Explain.
12. What is transaction? Draw states of transaction and explain.